

TABLE 10.1 Frequently used Taylor series

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots + x^n + \cdots = \sum_{n=0}^{\infty} x^n, \quad |x| < 1$$

$$\frac{1}{1+x} = 1 - x + x^2 - \cdots + (-x)^n + \cdots = \sum_{n=0}^{\infty} (-1)^n x^n, \quad |x| < 1$$

$$e^x = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad |x| < \infty$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad |x| < \infty$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + (-1)^n \frac{x^{2n}}{(2n)!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \quad |x| < \infty$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + (-1)^{n-1} \frac{x^n}{n} + \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n}, \quad -1 < x \leq 1$$

$$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots + (-1)^n \frac{x^{2n+1}}{2n+1} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}, \quad |x| \leq 1$$

The Binomial Series

For $-1 < x < 1$,

$$(1+x)^m = 1 + \sum_{k=1}^{\infty} \binom{m}{k} x^k,$$

where we define

$$\binom{m}{1} = m, \quad \binom{m}{2} = \frac{m(m-1)}{2!},$$

and

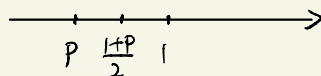
$$\binom{m}{k} = \frac{m(m-1)(m-2) \cdots (m-k+1)}{k!} \quad \text{for } k \geq 3.$$

Q1. 形如 $\sum_{n=1}^{\infty} \frac{1}{n^p (\ln n)^q (\ln \ln n)^r}$ 级数敛散性? 其中 $p, q, r \in \mathbb{R}$.

Sol: 先看 p :

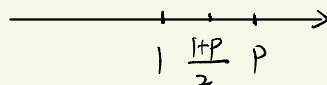
① $p < 1$ 级数发散

原因: 与 $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{1+p}{2}}}$ 比较.



② $p > 1$ 级数收敛

原因: 与 $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{1+p}{2}}}$ 比较

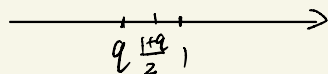


③ $p = 1$ 再看 q :

③.1 $q < 1$ 级数发散

原因: 与 $\sum_{n=1}^{\infty} \frac{1}{n (\ln n)^{\frac{1+q}{2}}}$ 比较,

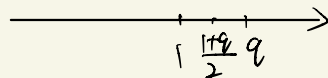
积分判别法证发散



③.2 $q > 1$ 级数收敛

原因: 与 $\sum_{n=1}^{\infty} \frac{1}{n (\ln n)^{\frac{1+q}{2}}}$ 比较

积分判别法证收敛

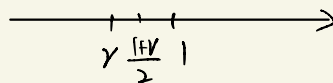


③.3 $q = 1$ 再看 r :

③.3.1 $r < 1$ 级数发散

原因: 与 $\sum_{n=1}^{\infty} \frac{1}{n \ln n (\ln \ln n)^{\frac{1+r}{2}}}$ 比较

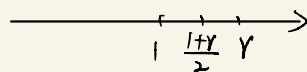
积分判别法证发散



③.3.2 $r > 1$ 级数收敛

原因: 与 $\sum_{n=1}^{\infty} \frac{1}{n \ln n (\ln \ln n)^{\frac{1+r}{2}}}$ 比较

积分判别法证收敛



③.3.3 $r = 1$ 级数发散

原因: 积分判别法

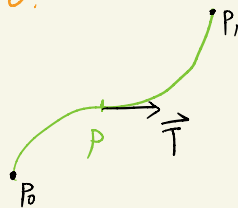
□

Q2. 参数曲线 $\vec{r}(t)$, 弧长参数 $s(t)$, 切向量 $\vec{T}$, 单位切向量 $\vec{T}$
曲率 k , 单位主法向量, 曲率圆 (密切圆)?

Sol: 一般的参数曲线 $\vec{r}(t) = (x(t), y(t), z(t)) \quad t \in [t_0, t_1]$

注意: 对于形如 $y=f(x)$ 的平面曲线可以将其参数化:

$$\vec{r}(t) = (x(t), y(t)), \text{ 其中 } \begin{cases} x(t) = t \\ y(t) = f(t) \end{cases}$$



切向量 $\vec{v} = \frac{d\vec{r}}{dt} = (x'(t), y'(t), z'(t))$

单位切向量:

$$\vec{T} = \frac{\vec{v}}{|\vec{v}|} = \left(\frac{x'(t)}{\sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2}}, \frac{y'(t)}{\sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2}}, \frac{z'(t)}{\sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2}} \right)$$

弧长参数: $s(t) = \int_{t_0}^t \sqrt{[x'(\tau)]^2 + [y'(\tau)]^2 + [z'(\tau)]^2} d\tau = \int_{t_0}^t |\vec{v}(\tau)| d\tau$

可以看出 $\frac{ds}{dt} = \sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2} = |\vec{v}(t)| = \left| \frac{d\vec{r}}{dt} \right|$

如果可以基于 $s(t)$ 反过来用 s 表达 t (比如 $s(t) = 13t$, 则 $t = \frac{s}{13}$)

那么可以把 $\vec{r}(t)$ 中的 t 用 s 替换 (比如此时 $\vec{r}(t) = \vec{r}(\frac{s}{13})$)

那么称曲线 $\vec{r}(s)$ 为基于弧长参数 s 的曲线

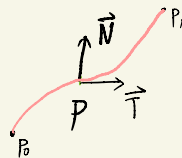
$$= (x(\frac{s}{13}), y(\frac{s}{13}), z(\frac{s}{13}))$$

下面为参数是弧长参数时的曲率计算。

此时 $\vec{T} = \frac{d\vec{r}}{dt} / \left| \frac{d\vec{r}}{dt} \right| = \frac{d\vec{r}}{dt} / \frac{ds}{dt} = \frac{d\vec{r}}{ds}$

曲率 $k := \left| \frac{d\vec{T}}{ds} \right| = \left| \frac{d^2\vec{r}}{ds^2} \right|$ 就是二阶导的模长, 是取值为正的关于 s 的函数!!!

单位主法向量 $\vec{N} := \frac{d\vec{T}}{ds} / \left| \frac{d\vec{T}}{ds} \right| = \frac{1}{k} \frac{d\vec{T}}{ds} = \frac{1}{k} \frac{d^2\vec{r}}{ds^2}$



下面为参数 t 不一定是弧长参数时的曲率计算。

$$\kappa := \left| \frac{d\vec{T}}{ds} \right| = \left| \frac{d\vec{T}}{dt} \cdot \frac{dt}{ds} \right| = \frac{\left| \frac{d\vec{T}}{dt} \right|}{\left| \frac{ds}{dt} \right|} = \frac{1}{|\vec{v}|} \left| \frac{d\vec{T}}{dt} \right| = \left(\frac{1}{|\vec{v}|} \right) \left| \frac{d\vec{T}}{dt} \right|$$

$$\vec{N} := \frac{d\vec{T}/ds}{|d\vec{T}/ds|} = \frac{(d\vec{T}/dt)(dt/ds)}{|d\vec{T}/dt| |dt/ds|} = \frac{d\vec{T}/dt}{|d\vec{T}/dt|}$$

$$\therefore \frac{dt}{ds} > 0.$$

与上面情况形式一致。

比上面情况多了这个。

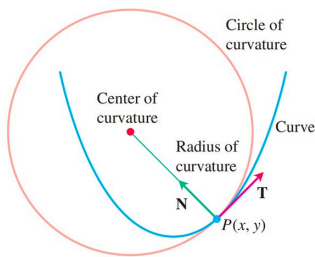


FIGURE 13.20 The center of the osculating circle at $P(x, y)$ lies toward the inner side of the curve.

Circle of Curvature for Plane Curves

The **circle of curvature** or **osculating circle** at a point P on a plane curve where $\kappa \neq 0$ is the circle in the plane of the curve that

1. is tangent to the curve at P (has the same tangent line the curve has)
2. has the same curvature the curve has at P
3. has center that lies toward the concave or inner side of the curve (as in Figure 13.20).

The **radius of curvature** of the curve at P is the radius of the circle of curvature, which, according to Example 2, is

曲率圆的半径是曲率的倒数!

$$\text{Radius of curvature} = \rho = \frac{1}{\kappa}.$$

To find ρ , we find κ and take the reciprocal. The **center of curvature** of the curve at P is the center of the circle of curvature.